Lucas Swierad

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Education

University of Michigan — Ann Arbor, MI
Bachelor of Science in Computer Science & Bachelor of Science in Robotics — *Aug. 2023 – May 2027*

Experience

Software Engineering Intern — Sep. 2026 – Present
Michigan Kidney Translational Medicine Center (MiKTMC) — *Ann Arbor, MI*

- Self-hosted and configured a Linux server to run the Arvados platform, authoring YAML-based configuration files to define cluster, service, and deployment settings for a biomedical data infrastructure designed to scale to 1,000+ terabytes
- Developed and maintained automated Python and Bash scripts to streamline biomedical data ingestion, processing, and ETL workflows across 100+ terabytes of genomic and clinical datasets

Software Engineering Fellowship — Jan. 2026 – Present
UM CAEN — *Ann Arbor, MI*

- Developed a modular full-stack workflow automation platform (React + Node.js) built for 33,000+ faculty and staff, integrating calendar, file, and notification workflows
- Built backend services (FastAPI, SQLModel) for a Google Workspace storage-reclamation application, implementing OAuth 2.0 authentication restricted to @umich.edu accounts and a Drive indexing service backed by Redis caching and PostgreSQL
- Containerized the backend with Docker and configured CI/deployment to AWS App Runner, managing production environment configuration and enforcing secure secret handling

Site Reliability Engineering Intern — June 2026 – Aug. 2026
Barracuda Networks — *Ann Arbor, MI*

- Migrated 14 active CI/CD pipelines from Jenkins to GitHub Actions, enabling retirement of the organization's legacy Jenkins infrastructure
- Hardened Azure deployment workflows by using Terraform to provision managed identities and Azure Key Vault access for runtime secret retrieval, eliminating persistent credentials from CI/CD configuration

Projects

Robot Task Platform | *Java, Spring Boot, DynamoDB, gRPC, Docker, K8s, EKS, Terraform* — Sep. 2026

- Developed an event-driven hardware-in-the-loop testing platform across isolated Kubernetes jobs on AWS EKS
- Built Java/Spring Boot control-plane services using REST, gRPC, PostgreSQL, and Amazon EventBridge/SQS
- Created a Next.js and TypeScript operator console for launching tests, monitoring telemetry, investigating failures, and approving production certification through OAuth 2.0/OIDC role-based access controls
- Provisioned EKS, RDS PostgreSQL, DynamoDB, IAM, networking, ECR, CodeArtifact, and observability resources using Terraform; configured simulated hardware bridge nodes using Ansible
- Established GitHub Actions CI/CD pipelines for Gradle and npm builds, automated JUnit/Testcontainers/Cucumber/Cypress testing, container security scanning, artifact publishing, and Kubernetes deployment

AI-Powered Bug Tracker | *React.js, FastAPI, MongoDB, Gemini, Python, JavaScript* — Jan. 2026

- Built a full-stack web app with a React.js frontend for data entry, filtering, and visualization, backed by a FastAPI REST backend handling authentication and structured data retrieval
- Integrated Google Gemini to deliver AI-assisted insights, letting users query and summarize historical data to surface recurring bugs, learning patterns, and technical growth

Technical Skills

Programming Languages: Python, C++, C, Java, MATLAB, Julia, JavaScript, TypeScript, SQL
Frameworks: FastAPI, Flask, React, Node.js, Next.js, JUnit, WordPress, Cypress, Lint,
Developer Tools: GIT/GitHub, Linux, Azure, AWS, Kubernetes, Docker, CI/CD (Jenkins + GitHub Actions), REST APIs, Postgres, MongoDB, Terraform, Jira, Ansible, VS Code, Agile Development, Bash, Shell
Systems & Middleware: ROS 2, LCM, Raspberry Pi, RP2040, Gazebo, TCP/IP Socket, Serialization
Relevant Coursework: Data Structures, Algorithms, System Design, Operating Systems, Database Management